Prof. Truong-Son Hy

Website: `https://hytruongson.github.io/HySonLab/` Email: thy@uab.edu Office: (+1) 205-934-8604

Research Interest **HySonLab** is my research group. We work on graph neural networks, deep generative models on graphs, equivariant neural networks, multiresolution machine learning in the direction of **AI for Science, Bioinformatics, Drug Discovery, Drug Repurposing, and Medical AI**.

Education

The University of Chicago September 2016 - June 2022
Illinois, USA
PhD. Computer Science
Thesis: Graph representation learning, deep generative models on graphs, group equivariant molecular neural networks and multiresolution machine learning.
Advisor: Prof. Risi Kondor

The University of Chicago September 2016 - December 2018
Illinois, USA
MSc. Computer Science
Thesis: Covariant compositional networks for learning graphs and GraphFlow deep learning framework in C++/CUDA.
Advisor: Prof. Risi Kondor

Eötvös Loránd University September 2013 - July 2016
Budapest, Hungary
BSc. Computer Science
Thesis: Semi-supervised Adaptive Facial Tracking Method.
Advisor: Prof. Lőrincz András
GPA: **4.98/5** (in Hungary scale), **3.984/4** (in US scale)
Award: First-class Graduation Honour
Sponsor: Stipendium Hungaricum Full Scholarship from the Government of Hungary

Current Position

University of Alabama at Birmingham **(Tenure-Track) Assistant Professor**
Department of Computer Science August 2024 – present

- **Inaugural Phyllis and David Brasfield Endowed Faculty Scholar in Computer Science (Oct. 2026 - Sep. 2028)**
- Affiliated faculty with Heersink School of Medicine (UAB), Center for Clinical and Translational Science (CCTS).
- Research and advise PhD students in AI for Science, Bioinformatics, Drug Discovery & Medical AI.
- UAB Scholars: `https://scholars.uab.edu/19454-truongson-hy/`
- Faculty Profile: `https://www.uab.edu/cas/computerscience/people/faculty-directory/faculty/hy-truong-son`
- Teach CS 660/760 (MSc/PhD) – Artificial Intelligence: Falls 2024–2026 [Syllabus].
- Teach CS 663/763 (MSc/PhD) – Data Mining: Springs 2025–2026 [Syllabus].
- Teach CS 355/555 (BSc/MSc) – Probability & Statistics: Falls 2025–2026 [Syllabus].

Work Experience

Indiana State University **(Tenure-Track) Assistant Professor**
Department of Mathematics and Computer Science August 2023 – July 2024

- Fall 2023: CS 617 – Data Mining [Syllabus]
- Spring 2024: CS 417 – Machine Learning, CS 475/575 – Artificial Intelligence
- Serve in faculty search committees for Departments of Biology and Computer Science.

Porter Cancer Research Center **Affiliated Faculty**
Terre Haute, Indiana February 2024 – July 2024

- Deep Learning for Drug Discovery & Drug Repurposing.

The Center for Genomic Advocacy **Affiliated Faculty**
Terre Haute, Indiana February 2024 – July 2024

- Genomic Large Language Models.

University of California San Diego **Postdoctoral Fellow & Lecturer**
Halicioglu Data Science Institute September 2022 – April 2023

- Mentor: Prof. Yusu Wang.
- I taught DSC 40A - Theoretical Foundations of Data Science (Fall 2022) [Course Materials].

Facebook Inc. **PhD Intern**
Ads Machine Learning, Seattle office June – August 2020
Project:

- Task: Implementation of a transfer learning scheme in distributed system for large-scale sparse neural networks to classify and rank two billion users' advertisements. The framework is built based on Caffe2.
- Programming language: Python.
- Result: The models are implemented, trained and tested on 1 week of Facebook data.

Google Inc. **Site Reliability Engineer Intern**
Networking SRE, Sunnyvale office June – September 2018, 2019
Project:

- Task: Utilize machine learning and statistical techniques including message passing neural networks to design a methodology for root cause analysis and auto detection of network failures based on network events and topology. The Google's technologies used for this project include Unified Network Model, Borgmon and TensorFlow.
- Programming language: C++ and Python.
- Result: Implementation of several Machine Learning methods on real production data of network topologies and monitoring systems.

Google Inc. **Software Engineer Intern**
Security and Privacy, Chicago office June 2017 – September 2017
Project:

- Task: Analyze series of Googlers' accesses of sensitive data to detect anomalous behaviors from internal actors. Replace the daily analysis by a real-time analysis to reduce the latency of the current continuous pipeline. Apply Machine Learning and statistical models to evaluate and cluster data access logs in Google distributed systems. The challenge is to integrate with the existing Google's infrastructure, learn and use multiple Google's technologies such as Bigtable and Flume.
- Programming language: Java.
- Result: Successfully tested with artificially generated data that simulates the working pipeline in one year.

Neural Information Processing Group **Undergraduate Research Student**
Eötvös Loránd University April 2014 – July 2016

- Website: `http://nipg04.inf.elte.hu/`
- Working under the supervision of Prof. Lőrincz András with main focus on facial analysis and sparse coding learning algorithms.

Awards

Golden Globe Science & Technology Award November 17, 2026
Recipient of the 2026 Golden Globe Science & Technology Award in the Digital Technology category. A prestigious national award jointly presented by Vietnam's Ministry of Science and Technology and the Central Committee of the Ho Chi Minh Communist Youth Union, recognizing outstanding achievements and contributions in science and technology. Newspapers:

- Tien Phong: [link]
- Thanh Nien: [link]
- Bao Moi: [link]
- The Thao & Van Hoa: [link]

Phyllis and David Brasfield Endowed Faculty Scholarship 2026 – 2028
Selected as the Phyllis and David Brasfield Endowed Faculty Scholar in Computer Science in recognition of scholarly accomplishments, continued research excellence, and contributions to the Department of Computer Science and University of Alabama at Birmingham.

Vietnam International Awards (ViAwards) [Certificate] [website] November 8, 2025
One of 10 worldwide recipients of the 2025 Vietnam International Awards in London, United Kingdom, honored for outstanding contributions in Research, Innovation, and Artificial Intelligence. Newspapers:

- The World and Vietnam Report: [link]
- Vietnam News: [link]

Data Science Postdoctoral Fellowship September 2022 – July 2024
Awarded by the Halicioglu Data Science Institute at University of California San Diego.

Postdoctoral Fellowship (Declined) September 2022 – July 2024
Awarded by the Vector Institute (Canada).

University Unrestricted (UU) PhD Fellowship Spring quarter 2022
Awarded by the graduate committee at University of Chicago.

University Unrestricted (UU) PhD Fellowship Autumn quarter 2017
Awarded by the graduate committee at University of Chicago.

First-class Graduation Honor Class of 2016
Awarded by the highest GPA achiever by Faculty of Informatics at Eötvös Loránd University [Certificate].

The title of Excellent Student of the Faculty Academic year 2014 - 2015
Awarded to 2 out of 4,000 BSc. students (0.05%) of the Faculty of Informatics with outstanding academic performance and scientific activity at Eötvös Loránd University [Certificate].

Stipendium Hungaricum Full Scholarship September 2013 – July 2016
Awarded by the Government of Hungary that covers tuition fee and living expenses.

First position at National Conference of Students' Scientific Association April 2015
Thesis: *Semi-supervised Adaptive Facial Tracking Method.* [Certificate]

Morgan Stanley Scholarship 2015
Amount: 75,000 HUF.

1st place at Hungarian ACM Programming Contest October 2015
Team ELTE-Sparrows solved 9/10 problems, ranked 1 out of 35 teams nationwide [Certificate].

The 2015 ACM ICPC Central Europe Regional Contest November 2015
Team ELTE-1 achieved Honorable Mention [Certificate]. Location: Zagreb, Croatia.

2nd place at Hungarian ACM Programming Contest October 2014
Team ELTE-UFMG-UFPB solved 5/10 problems, ranked 2 out of 33 teams nationwide [Certificate].

The 2014 ACM ICPC Central Europe Regional Contest November 2014
Team ELTE-2 achieved Honorable Mention [Certificate]. Location: Krakow, Poland .

7th place at ACM ICPC Asia Regional Programming Contest November 2012
Team DiscreteMath achieved Consolidation prize, ranked 7 out of 59 teams from Asia [Certificate]. Location: Haiyai, Thailand.

Special prize at the NAPROCK 4th International Programming Contest 2012
[Certificate]. Location: Omuta, Japan.

Silver Cup at the Vietnam Olympic of University Students in Informatics 2012
[Certificate]. Location: Hanoi, Vietnam.

Runner-up prize at the National Excellent Student Contest in Informatics 2011
[Certificate]. Location: Hanoi, Vietnam.

Conference Proceedings [Google Scholar] [ORCID]

[C26] Quy-Anh Dang, Chris Ngo, and **Truong-Son Hy** [c], *RainbowPlus: Enhancing Adversarial Prompt Generation via Evolutionary Quality-Diversity Search*, **EMNLP 2026**.
c: Corresponding author / PI.
https://aclanthology.org/2026.emnlp-industry.70.pdf

[C25] Dinh-Khanh Pham, Quy-Anh Dang, Lam Mai Thanh, Khanh Bui, and **Truong-Son Hy** [c], *From SQL to Knowledge Graphs: An LLM-Driven Multi-Agent Approach with Data Schema Improvement*, **2026 IEEE International Conference on Information Reuse and Integration for Data Science (IRI)**, DOI 10.1109/IRI69576.2026.00094.
c: Corresponding author / PI.
https://ieeexplore.ieee.org/document/11676779 (official proceedings)
https://imyday.github.io/pub/iri2026/pdfs/8431-3vA8WWeLJ21imKnq5mvHBx/843101a478/843101a478.pdf (temporary proceedings)

[C24] Ngoc-Son Nguyen, Hieu-Nghia Huynh-Nguyen, Thanh V. T. Tran, **Truong-Son Hy**, and Van Nguyen, *DiFlow-TTS: Discrete Flow Matching with Factorized Speech Tokens for Low-Latency Zero-Shot Text-To-Speech*, **Interspeech 2026**.
https://arxiv.org/pdf/2509.09631.pdf

[C23] Tuan-Anh Yang, Bao V. Q. Bui, Chanh-Quang Vo-Van, **Truong-Son Hy** [c], *Halfway to 3D: Ensembling 2.5D and 3D Models for Robust COVID-19 CT Diagnosis*, **Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops**, 2026, pp. 6459-6465.
https://openaccess.thecvf.com/content/CVPR2026W/PHAROS-AIF-MIH/papers/Yang_Halfway_to_3D_Ensembling_2.5D_and_3D_Models_for_Robust_CVPRW_2026_paper.pdf

[C22] Ngoc-Son Nguyen, Thanh V. T. Tran, Jeongsoo Choi, Hieu-Nghia Huynh-Nguyen, **Truong-Son Hy**, and Van Nguyen, *DiFlowDubber: Discrete Flow Matching for Automated Video Dubbing via Cross-Modal Alignment and Synchronization*, DOI 10.1109/CVPRF71909.2026.00561, **CVPR 2026** (Findings).
https://ieeexplore.ieee.org/document/11697835

[C21] Duy Do, **Truong-Son Hy** [c], and Thang Dinh, *RELIC: Reinforcement Learning Based Ising Optimization via Graph Compression*, **2025 IEEE International Conference on Quantum Computing and Engineering (QCE)**, Albuquerque, NM, USA, 2025, pp. 193-198, DOI 10.1109/QCE65121.2025.10318.
c: co-PI.
https://ieeexplore.ieee.org/document/11250082

[C20] Tan-Hanh Pham, Hoang-Nam Le, Phu-Vinh Nguyen, Chris Ngo, and **Truong-Son Hy** [c], *SilVar: Speech Driven Multimodal Model for Reasoning Visual Question Answering and Object Localization*, **EMNLP 2025**.
c: Corresponding author / PI.
https://aclanthology.org/2025.emnlp-main.589.pdf

[C19] Tan-Hanh Pham, Bui Trong Duong, Luu Quang Minh, Pham Tan Huong, Chris Ngo, and **Truong-Son Hy** [c], *SilVar-Med: A Speech-Driven Visual Language Model for Explainable Abnormality Detection in Medical Imaging*, **Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops**, 2025, pp. 3009-3019. .
https://openaccess.thecvf.com/content/CVPR2025W/MAR/papers/Pham_SilVar-Med_A_Speech-Driven_Visual_Language_Model_for_Explainable_Abnormality_Detection_CVPRW_2025_paper.pdf

[C18] Nghia Huynh Nguyen Hieu, Ngoc Son Nguyen, Huynh Nguyen Dang, Thieu Vo, **Truong-Son Hy**, and Van Nguyen, *OZSpeech: One-step Zero-shot Speech Synthesis with Learned-Prior-Conditioned Flow Matching*, **ACL 2025**.
https://aclanthology.org/2025.acl-long.1043.pdf

[C17] Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, Bach Phan Tat, Minh-Huong Ngo, Thanh Nguyen-Tang, **Truong-Son Hy** [c], *MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder*, **ACL 2025**.

c: Corresponding author / PI.
https://aclanthology.org/2025.acl-industry.79.pdf

[**C16**] Khai-Nguyen Nguyen, Khai Le-Duc, Bach Phan Tat, Le Duy, Long Vo-Dang, **Truong-Son Hy** [c], *Sentiment Reasoning for Healthcare*, **ACL 2025**.
c: Corresponding author / PI.
https://aclanthology.org/2025.acl-industry.82.pdf

[**C15**] Khai Le-Duc, David Thulke, Hung-Phong Tran, Long Vo-Dang, Khai-Nguyen Nguyen, **Truong-Son Hy**, and Ralf Schluter, *Medical Spoken Named Entity Recognition*, **NAACL 2025**.
https://aclanthology.org/2025.naacl-industry.59.pdf

[**C14**] Cuong Tran Van, Thanh V. T. Tran, Van Nguyen, and **Truong-Son Hy** [c], *Effective Context Modeling Framework for Emotion Recognition in Conversations*, **ICASSP 2025**, DOI 10.1109/ICASSP49660.2025.10888112.
c: Corresponding author / PI.
https://ieeexplore.ieee.org/document/10888112

[**C13**] Co Tran, Quoc-Bao Tran, **Truong-Son Hy** [c], and Thang N. Dinh, *Scalable Quantum-Inspired Optimization through Dynamic Qubit Compression*, **AAAI 2025** (Oral Presentation).
c: co-PI.
https://ojs.aaai.org/index.php/AAAI/article/view/33235/35390

[**C12**] Thanh V. T. Tran, Nhat Khang Ngo, Viet Anh Nguyen, and **Truong-Son Hy** [c], *GROOT: Effective Design of Biological Sequences with Limited Experimental Data*, **KDD 2025**, Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1., Pages 1385-1396, DOI 10.1145/3690624.3709291. .
c: Corresponding author / PI.
https://dl.acm.org/doi/10.1145/3690624.3709291 (paper)
https://arxiv.org/pdf/2411.11265.pdf (preprint)

[**C11**] Khai Le-Duc, Khai-Nguyen Nguyen, Long Vo-Dang, and **Truong-Son Hy** [c], *Real-time Speech Summarization for Medical Conversations*, **Interspeech 2024**, DOI 10.21437/Interspeech.2024-2250.
c: Corresponding author / PI.
https://www.isca-archive.org/interspeech_2024/leduc24_interspeech.pdf

[**C10**] Zhishang Luo, **Truong-Son Hy**, Puoya Tabaghi, Michael Defferrard, Elahe Rezaei, Ryan Carey, Rhett Davis, Rajeev Jain, and Yusu Wang, *DE-HNN: An effective neural model for Circuit Netlist representation*, **AISTATS 2024**, Proceedings of The 27th International Conference on Artificial Intelligence and Statistics, PMLR 238:4258-4266, 2024.
https://proceedings.mlr.press/v238/luo24a/luo24a.pdf

[**C9**] Thuan Trang, Nhat Khang Ngo, Daniel Levy, Thieu N. Vo, Siamak Ravanbakhsh, and **Truong-Son Hy** [c], *E(3)-Equivariant Mesh Neural Networks*, **AISTATS 2024**, Proceedings of The 27th International Conference on Artificial Intelligence and Statistics, PMLR 238:748-756, 2024.
c: Corresponding author / PI.
https://proceedings.mlr.press/v238/anh-trang24a/anh-trang24a.pdf

[**C8**] Minh H. Nguyen, Nghi D. Q. Bui, **Truong-Son Hy**, Long Tran-Thanh, and Tien N. Nguyen, *HierarchyNet: Learning to Summarize Source Code with Heterogeneous Representations*, **EACL 2024**.
https://aclanthology.org/2024.findings-eacl.156.pdf

[**C7**] Duc Thien Nguyen*, Manh Duc Tuan Nguyen*, **Truong-Son Hy** *[c], and Risi Kondor, *Fast Temporal Wavelet Graph Neural Networks*, **NeurIPS 2023** (Workshop on Symmetry and Geometry in Neural Representations), Proceedings of Machine Learning Research 228:35-54.
∗: Co-first authors.
c: Corresponding author / PI.
https://proceedings.mlr.press/v228/nguyen24a.html
https://openreview.net/pdf?id=Mo5qZaBl8v

[**C6**] Chen Cai, **Truong-Son Hy**, Rose Yu, and Yusu Wang, *On the Connection Between MPNN and Graph Transformer*, **ICML 2023**, Proceedings of Machine Learning Research 202:3408-3430.
`https://proceedings.mlr.press/v202/cai23b/cai23b.pdf`

[**C5**] Cong Dao Tran, Nhut Huy Pham, Anh Nguyen, **Truong-Son Hy** [c], and Tu Vu, *ViDeBERTa: A powerful pre-trained language model for Vietnamese*, **EACL 2023**.
c: Corresponding author / PI.
`https://aclanthology.org/2023.findings-eacl.79.pdf`

[**C4**] **Truong-Son Hy**, Viet Bach Nguyen, Long Tran-Thanh and Risi Kondor, *Temporal Multiresolution Graph Neural Networks For Epidemic Prediction*, **ICML 2022** (Workshop on Healthcare AI and COVID-19), Proceedings of Machine Learning Research 184:21-32.
`https://proceedings.mlr.press/v184/hy22a/hy22a.pdf`

[**C3**] **Truong-Son Hy** and Risi Kondor, *Multiresolution Matrix Factorization and Wavelet Networks on Graphs*, **ICML 2022** (Workshop on Topology, Algebra, and Geometry in Machine Learning), Proceedings of Machine Learning Research 196:172-182.
`https://proceedings.mlr.press/v196/hy22a/hy22a.pdf`
`https://arxiv.org/pdf/2111.01940.pdf` (long version)

[**C2**] Brandon Anderson, **Truong-Son Hy** and Risi Kondor. *Cormorant: Covariant molecular neural networks*, **NeurIPS 2019**.
`https://dl.acm.org/doi/pdf/10.5555/3454287.3455589`

[**C1**] Risi Kondor, **Truong-Son Hy**, Horace Pan, Brandon Anderson and Shubhendu Trivedi, *Covariant compositional networks for learning graphs*, **ICLR 2018**.
`https://arxiv.org/pdf/1801.02144.pdf`

Journal Publications
[Google Scholar]
[ORCID]

[**J28**] Phuc Pham and **Truong-Son Hy** [c], *Evaluating Agentic Bioinformatics through Function, Evidence, and Validation*, accepted **Briefings in Bioinformatics** (Q1, Impact Factor = 7.3), Year 2026.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2607.27556.pdf`

[**J27**] Rebin Saleh, Khanh Pham Dinh, Balazs Villanyi, and **Truong-Son Hy** [c], *HAMA: A Hierarchical Adaptive Multi-Agent Architecture for Industrial IoT Predictive Maintenance*, **IEEE Access**, Volume 14, Pages 142595-142613, Year 2026, DOI 10.1109/ACCESS.2026.3731096.
c: Corresponding author / PI.
`https://ieeexplore.ieee.org/document/11678972`

[**J26**] Phuc Pham, Viet Thanh Duy Nguyen, Kevin Song, Jake Chen and **Truong-Son Hy** [c], *LINKER: Learning Interactions Between Functional Groups and Residues With Chemical Knowledge-Enhanced Reasoning and Explainability*, **Journal of Chemical Information and Modeling** (Q1, Impact Factor = 6.4), Volume 66, Number 14, Pages 8579-8593, Year 2026, Month 07, DOI 10.1021/acs.jcim.6c00527.
c: Corresponding author / PI.
`https://pubs.acs.org/doi/10.1021/acs.jcim.6c00527`

[**J25**] Minh Ngoc Nguyen, Khai Le-Duc, Tan-Hanh Pham, Trong Nhan Nguyen, Trang Nguyen, Ba Kien Tran, Viktor Dremin, Sergei Sokolovsky, Edik Rafailov, and **Truong-Son Hy** [c], *A Wearable Device Dataset for Mental Health Assessment Using Laser Doppler Flowmetry and Fluorescence Spectroscopy Sensors*, **Nature - Communications Medicine** (Q1, Impact Factor = 7.4), Volume 6, Article Number 561, Year 2026, DOI 10.1038/s43856-026-01766-5.
c: Corresponding author / PI.
`https://www.nature.com/articles/s43856-026-01766-5`

[**J24**] Cuong Manh Nguyen and **Truong-Son Hy** [c], *A comprehensive review of efficient deep learning for clinical medical imaging deployment*, **Discover Artificial Intelligence** (Q1), Volume 6, Article Number 1252, Year 2026, DOI 10.1007/s44163-026-01706-1.
c: Corresponding author / PI.
`https://link.springer.com/article/10.1007/s44163-026-01706-1`

[**J23**] Luu Tung Hai, Thinh D. Le, Zhicheng Ding, Qing Tian, and **Truong-Son Hy** [c], *Topology-Guided Knowledge Distillation for Efficient Point Cloud Processing*, **Transactions on Machine Learning Research (TMLR)**, Year 2026.
c: Corresponding author / PI.
`https://openreview.net/pdf?id=lP2phGa5af`

[**J22**] **Truong-Son Hy**, Thieu Khang Nguyen, and Risi Kondor, *Learning to Solve Multiresolution Matrix Factorization by Manifold Optimization and Evolutionary Metaheuristics*, **Machine Learning: Science and Technology** (Q1, Impact Factor = 4.6), Volume 7, Number 2, Pages 025060, Year 2026, DOI 10.1088/2632-2153/ae5b22.
`https://iopscience.iop.org/article/10.1088/2632-2153/ae5b22`

[**J21**] Tien Dang [*] and **Truong-Son Hy** [*c], *EquiHGNN: Scalable Rotationally Equivariant Hypergraph Neural Networks*, **Journal of Chemical Physics** (Q1, Impact Factor = 3.1), Volume 164, Number 14, Pages 1444101, Month 04, Year 2026, DOI 10.1063/5.0317966.
∗: Co-first authors.
c: Corresponding author / PI.
`https://doi.org/10.1063/5.0317966`

Presented at **NeurIPS 2025** workshops:

- AI for Science: `https://openreview.net/pdf?id=McZIwV1dva`
- Machine Learning and the Physical Sciences: `https://openreview.net/pdf?id=NapJ4sLtSb`
- New Perspectives in Graph Machine Learning: `https://openreview.net/pdf?id=2YIhVthm0F`

[**J20**] Quang Dung Dinh, Daniel Kunk, **Truong-Son Hy** [c], Vamsi J Nalam, and Phuong Dao, *DiscoEPG: A Python package for characterization of insect electrical penetration graph (EPG) signals*, Smart Agricultural Technology (Q1, Impact Factor = 5.7), Volume 13, Page 101874, Year 2026, DOI 10.1016/j.atech.2026.101874.
c: Corresponding author / PI.
`https://www.sciencedirect.com/science/article/pii/S2772375526000985`

[**J19**] Viet Thanh Duy Nguyen, Phuc Pham, and **Truong-Son Hy** [c], *EVOSYNTH: Enabling Multi-Target Drug Discovery through Latent Evolutionary Optimization and Synthesis-Aware Prioritization*, **Nature - Communications Chemistry** (Q1, Impact Factor = 5.9), Volume 9, Number 133, Year 2026, DOI 10.1038/s42004-026-01945-4.
c: Corresponding author / PI.
`https://www.nature.com/articles/s42004-026-01945-4`

[**J18**] Tien Dang, Viet Thanh Duy Nguyen, Minh Tuan Le, and **Truong-Son Hy** [c], *BioMedKG: Multimodal Contrastive Representation Learning in Augmented Biomedical Knowledge Graphs*, **Frontiers in Systems Biology** (Q1, Impact Factor = 2.3), DOI 10.3389/fsysb.2025.1651930, Volume 5, Year 2025.
c: Corresponding author / PI.
`https://www.frontiersin.org/journals/systems-biology/articles/10.3389/fsysb.2025.1651930/full`

[**J17**] Viet Anh Nguyen, Nhat Khang Ngo, and **Truong-Son Hy** [c], *Range-aware Positional Encoding via High-order Pretraining: Theory and Practice*, **Machine Learning: Science and Technology** (Q1, Impact Factor = 4.6), Volume 6, Number 4, Year 2025, DOI 10.1088/2632-2153/ae1acd.
c: Corresponding author / PI.
`https://iopscience.iop.org/article/10.1088/2632-2153/ae1acd`

[**J16**] Quang P. M. Pham, Khoi T. N. Nguyen, Lan C. Ngo, Dezhen Song, Truong Do, and **Truong-Son Hy** [c], *TESGNN: Temporal Equivariant Scene Graph Neural Networks for Efficient and Robust Multi-View 3D Scene Understanding*, accepted at **Transactions on Machine Learning Research (TMLR)**, Year 2025.
c: Corresponding author / PI.
`https://openreview.net/pdf?id=boM0kkYPzE` (paper)
`https://arxiv.org/pdf/2411.10509.pdf` (preprint)

[**J15**] Viet Tien Pham, Minh Hieu Ha, Bao V. Q. Bui and **Truong-Son Hy** [c], *FFTMed: Leveraging Fast-Fourier Transform for a Lightweight and Adversarial-Resilient Medical Image Segmentation Framework*, **Nature - Scientific Reports** (Q1, Impact Factor = 3.9, H-index = 347), Volume 15, Number 37879, Year 2025, DOI 10.1038/s41598-025-21799-5.
c: Corresponding author / PI.
`https://www.nature.com/articles/s41598-025-21799-5`

[**J14**] Tam Trinh, Anh Dao, Hy Thi Hong Nhung, and **Truong-Son Hy** [c], *VietMedKG: Knowledge Graph and Benchmark for Traditional Vietnamese Medicine*, **ACM Transactions on Asian and Low-Resource Language Information Processing** (Impact Factor = 1.8), Volume 24, Issue 7, Article No.: 69, Pages 1-17, Year 2025, DOI 10.1145/3744740.
c: Corresponding author / PI.
`https://dl.acm.org/doi/10.1145/3744740`

[**J13**] Phuc Pham, Viet Thanh Duy Nguyen, Kyu Hong Cho, and **Truong-Son Hy** [c], *DrugPipe: Generative AI-assisted Virtual Screening Pipeline for Generalizable and Efficient Drug Repurposing*, **Biology Methods & Protocols** (Q1, Impact Factor = 2.5), Volume 10, Issue 1, Year 2025, DOI 10.1093/biomethods/bpaf038.
c: Corresponding author / PI.
`https://doi.org/10.1093/biomethods/bpaf038`

[**J12**] Thanh V. T. Tran, Nhat Khang Ngo, Viet Thanh Duy Nguyen, and **Truong-Son Hy** [c], *LatentDE: Latent-based Directed Evolution for Protein Sequence Design*, **Machine Learning: Science and Technology** (Q1, Impact Factor = 6.3), Volume 6, Number 1, Year 2025, DOI 10.1088/2632-2153/adc2e2.
c: Corresponding author / PI.
`https://iopscience.iop.org/article/10.1088/2632-2153/adc2e2/pdf`

[**J11**] Quang-Dung Dinh, Daniel Kunk, **Truong-Son Hy** [c], Vamsi J. Nalam, and Phuong Dao, *Machine Learning for Automated Electrical Penetration Graph Analysis of Aphid Feeding Behavior: Accelerating Research on Insect-Plant Interactions*, **PLOS ONE** (Q1, Impact Factor = 2.9, H-index = 435), Volume 20, Number 4, Pages 1-25, Year 2025, DOI 10.1371/journal.pone.0319484.
c: Corresponding author / PI.
`https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319484`

[**J10**] Viet Thanh Duy Nguyen, Nhan Nguyen, and **Truong-Son Hy** [c], *ProteinReDiff: Complex-based ligand-binding proteins redesign by equivariant diffusion-based generative models*, **Structural Dynamics** (Q1, Impact Factor = 2.8), Volume 11, Issue 6, Year 2024, DOI 10.1063/4.0000271.
c: Corresponding author / PI.
`https://doi.org/10.1063/4.0000271`

[**J9**] Khanh-Tung Tran, **Truong-Son Hy**, Lili Jiang, and Xuan-Son Vu, *MGLEP: Multimodal Graph Learning for Modeling Emerging Pandemics with Big Data*, **Nature - Scientific Reports** (Q1, Impact Factor = 3.8, H-Index = 315), Volume 14, Number 16377, Year 2024, DOI 10.1038/s41598-024-67146-y.
`https://www.nature.com/articles/s41598-024-67146-y`

[**J8**] Viet Thanh Duy Nguyen* and **Truong-Son Hy** *[c], *Multimodal Pretraining for Unsupervised Protein Representation Learning*, **Biology Methods & Protocols** (Q1, Impact Factor = 3.6), Volume 9, Issue 1, Year 2024, DOI 10.1093/biomethods/bpae043.
∗: Co-first authors.
c: Corresponding author / PI.
`https://doi.org/10.1093/biomethods/bpae043`

[**J7**] Trong Thanh Tran* and **Truong-Son Hy** *[c], *Protein Design by Directed Evolution Guided by Large Language Models*, **IEEE Transactions on Evolutionary Computation** (Q1, Impact Factor = 14.3), vol. 29, no. 2, pp. 418-428, April 2025, DOI 10.1109/TEVC.2024.3439690.
∗: Co-first authors.
c: Corresponding author / PI.
`https://ieeexplore.ieee.org/document/10628050`
`https://www.biorxiv.org/content/10.1101/2023.11.28.568945v3.full.pdf` (Preprint)

[**J6**] Nhat Khang Ngo* and **Truong-Son Hy** *[c], *Multimodal Protein Representation Learning and Target-aware Variational Auto-encoders for Protein-binding Ligand Generation*, **Machine Learning: Science and Technology** (Q1, Impact Factor = 6.8), Volume 5, Number 2, Year 2024, DOI 10.1088/2632-2153/ad3ee4.
∗: Co-first authors.
c: Corresponding author / PI.
`https://iopscience.iop.org/article/10.1088/2632-2153/ad3ee4`
`https://openreview.net/pdf?id=4k926QVVM4` (Presented at **NeurIPS 2023**)

[**J5**] Thuan Nguyen Anh Trang, Khang Nhat Ngo, Hugo Sonnery, Thieu Vo, Siamak Ravanbakhsh, **Truong-Son Hy** [c], *Scalable Hierarchical Self-Attention with Learnable Hierarchy for Long-Range Interactions*, **Transactions on Machine Learning Research (TMLR)**, Year 2024.
c: Corresponding author / PI.
`https://openreview.net/pdf?id=qH4YFMyhce`

[**J4**] Thong Bach, Anh Tong, **Truong-Son Hy**, Vu Nguyen, and Thanh Nguyen-Tang, *Long-Tailed Visual Recognition with Global Contrastive Learning and Prototype Learning*, **Transactions on Machine Learning Research (TMLR)**, Year 2023.
`https://openreview.net/pdf?id=xWrtiJwJj5`

[**J3**] Nhat Khang Ngo*, **Truong-Son Hy** *[c], and Risi Kondor, *Multiresolution Graph Transformers and Wavelet Positional Encoding for Learning Long-Range and Hierarchical Structures*, **The Journal of Chemical Physics** (Q1, Impact Factor = 4.4), Volume 159, Issue 3, Year 2023, DOI 10.1063/5.0152833.
∗: Co-first authors.
c: Corresponding author / PI.
`https://doi.org/10.1063/5.0152833`
`https://arxiv.org/pdf/2302.08647.pdf` (Spotlight presentation at **ICML 2023**)

[**J2**] **Truong-Son Hy** and Risi Kondor, *Multiresolution Equivariant Graph Variational Autoencoder*, **Machine Learning: Science and Technology** (Q1, Impact Factor = 6.8), Volume 4, Number 1, Year 2023, DOI 10.1088/2632-2153/acc0d8.
`https://iopscience.iop.org/article/10.1088/2632-2153/acc0d8`
`https://arxiv.org/pdf/2106.00967.pdf` (Spotlight presentation at **ICML 2022**, AI for Science Workshop)

[**J1**] **Truong-Son Hy**, Shubhendu Trivedi, Horace Pan, Brandon M. Anderson and Risi Kondor, *Predicting molecular properties with covariant compositional networks*, **The Journal of Chemical Physics** (Q1, Impact Factor = 4.4), Volume 148, Issue 24, Year 2018, DOI 10.1063/1.5024797.
`https://aip.scitation.org/doi/10.1063/1.5024797`

Workshop Paper Presentations [Google Scholar] [ORCID]

[**W14**] Viet Thanh Duy Nguyen, John K. Johnstone, and **Truong-Son Hy** [c], *PRIME: Protein Representation via Physics-Informed Multiscale Equivariant Hierarchies*, **BIOKDD 2026**.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2605.01625.pdf`

[**W13**] Quy-Anh Dang, Chris Ngo, **Truong-Son Hy** [c] and , *RedBench: A Universal Dataset for Comprehensive Red Teaming of Large Language Models*, **ICLR 2026** (Trustworthy AI Workshop).
c: Corresponding author / PI.
`https://arxiv.org/pdf/2601.03699.pdf`

[**W12**] Cuong Manh Nguyen, Huy-Hoang Do-Huu, Viet Thanh Duy Nguyen, and **Truong-Son Hy** [c], *TeBaAb: Text-Based Antigen-Conditioned Antibody Redesign via Directed Evolution*, **NeurIPS 2025** Workshops.
c: Corresponding author / PI.
`https://openreview.net/pdf?id=Imw5NGMgje` (AI for Science Workshop)
`https://openreview.net/pdf?id=6kvLcoNjjo` (Machine Learning and the Physical Sciences)

[**W11**] Hiep V. Dang, Bach Nguyen, Phong C.H. Nguyen, and **Truong-Son Hy** [c], *AdaptFNO: Adaptive Fourier Neural Operator with Dynamic Spectral Modes and Multiscale Learning for Climate Modeling*, **NeurIPS 2025** (Machine Learning and the Physical Sciences Workshop).

c: Corresponding author / PI.
https://openreview.net/pdf?id=9SvLwi7z2R

[**W10**] Phuc Pham, Viet Thanh Duy Nguyen, and **Truong-Son Hy** [c], *LINKER: Learning Interactions Between Functional Groups and Residues With Chemical Knowledge-Enhanced Reasoning and Explainability*, **NeurIPS 2025** Workshops.
c: Corresponding author / PI.
https://arxiv.org/pdf/2509.03425.pdf (preprint)
https://openreview.net/pdf?id=LsDdZUSVso (AI for Science)
https://openreview.net/pdf?id=En4Q41ZA3T (Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences)
https://openreview.net/pdf?id=LQjn0ziVgk (Machine Learning and the Physical Sciences)

[**W9**] Tien Dang, Viet Thanh Duy Nguyen, Minh Tuan Le, and **Truong-Son Hy** [c], *Multimodal Contrastive Representation Learning in Augmented Biomedical Knowledge Graphs*, **BIOKDD 2025**: 24th International Workshop on Data Mining in Bioinformatics.
c: Corresponding author / PI.
https://arxiv.org/pdf/2501.01644.pdf

[**W8**] Viet Anh Nguyen, Nhat Khang Ngo, and **Truong-Son Hy** [c], *Range-aware Positional Encoding via High-order Pretraining: Theory and Practice*, **NeurIPS 2024** (Workshop on Symmetry and Geometry in Neural Representations).
c: Corresponding author / PI.
https://openreview.net/pdf?id=tN0n5BuLEI (extended abstract)
https://arxiv.org/pdf/2409.19117

[**W7**] Thanh V. T. Tran, Nhat Khang Ngo, Viet Thanh Duy Nguyen, and **Truong-Son Hy** [c], *LatentDE: Latent-based Directed Evolution accelerated by Gradient Ascent for Protein Sequence Design*, **NeurIPS 2024** Workshops.
c: Corresponding author / PI.
https://openreview.net/pdf?id=4YkbQGVWGF (AI for Accelerated Materials Design)
https://openreview.net/pdf?id=8AmP6pQwyP (AI for New Drug Modalities)

[**W6**] Khai-Nguyen Nguyen, Khai Le-Duc, Bach Phan Tat, Le Duy, Jerry Ngo, Long Vo-Dang, Anh Totti Nguyen, and **Truong-Son Hy** [c], *Sentiment Reasoning for Healthcare*, **NeurIPS 2024** (Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond).
c: Corresponding author / PI.
https://openreview.net/pdf?id=WYn6YIjR41

[**W5**] Quang Pham Phuoc Minh, Tiet Nguyen Khoi Nguyen, Lan Chi Ngo, Tho Truong Do, and **Truong-Son Hy** [c], *ESGNN: Towards Equivariant Scene Graph Neural Network for 3D Scene Understanding*, **2024 33rd IEEE International Conference on Robot and Human Interactive Communication** (IEEE ROMAN).
c: Corresponding author / PI.
https://arxiv.org/pdf/2407.00609.pdf

[**W4**] Hugo Sonnery, Thuan Trang, Thieu N. Vo, Siamak Ravanbakhsh, and **Truong-Son Hy** [c], *Sequoia: Hierarchical Self-Attention Layer with Sparse Updates for Point Clouds and Long Sequences*, **ICLR 2023** (Workshop on Sparsity in Neural Networks).
c: Corresponding author / PI.
https://hytruongson.github.io/HySonLab/Sequoia_ICLR_2023.pdf

[**W3**] Viet Bach Nguyen*, **Truong-Son Hy***[c], Long Tran-Thanh, and Nhung Nghiem, *Predicting COVID-19 pandemic by spatio-temporal graph neural networks: A New Zealand's study*, **NeurIPS 2023** (Temporal Graph Learning Workshop).
∗: Co-first authors.
c: Corresponding author / PI.
https://openreview.net/pdf?id=tkjGiKs2g6
https://arxiv.org/pdf/2305.07731.pdf (long version)

[**W2**] Nhat Khang Ngo*, **Truong-Son Hy** *[c], and Risi Kondor, *Predicting Drug-Drug Interactions using Deep Generative Models on Graphs*, **NeurIPS 2022** (AI for Science Workshop).

∗: Co-first authors.
c: Corresponding author / PI.
https://arxiv.org/pdf/2209.09941.pdf

[**W1**] Yanan Long, Horace Pan, Chao Zhang, **Truong-Son Hy**, Risi Kondor, and Andrey Rzhetsky, *Molecular Fingerprints Are a Simple Yet Effective Solution to the Drug-Drug Interaction Problem*, **ICML 2022** (Workshop on Computational Biology).
https://icml-compbio.github.io/2022/papers/WCBICML2022_paper_72.pdf

Preprints & Under Review [Google Scholar] [ORCID]

[**P25**] Trong Nhan Nguyen, Vu Hoang Minh Doan, Thanh Dat Le, Truong Tien Vo, Thi Thu Ha Vu, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh, and **Truong-Son Hy** [c], *WaFIR: Wavelet-Fourier Mixture-of-Experts with Subband-Adaptive Routing for Blind All-in-One Image Restoration*, 2026.
c: Corresponding author / PI.
https://arxiv.org/pdf/2610.29108.pdf

[**P24**] Viet Thanh Duy Nguyen, John K. Johnstone, and **Truong-Son Hy** [c], *Towards a Unified Framework for Protein Interaction Site Prediction Using Hierarchical Protein Representations*, DOI 10.64898/2026.10.18.759950, 2026.
c: Corresponding author / PI.
https://www.biorxiv.org/content/10.64898/2026.10.18.759950v1.full.pdf

[**P23**] Tan-Phat Huynh, Dang-Manh Ho, Viet Thanh Duy Nguyen, Quang-Hien Kha, Ngoc-Thac Pham, Kuei-Hung Lai, and **Truong-Son Hy** [c], *MutaStruct: Revisiting Benchmarks for Single-Site Mutation-Induced Structural Change Prediction*, DOI 10.64898/2026.10.21.760837, 2026.
c: Corresponding author / PI.
https://www.biorxiv.org/content/10.64898/2026.10.21.760837v1.full.pdf

[**P22**] Thieu Khang Nguyen, Thu Huong Dang, and **Truong-Son Hy** [c], *A Hybrid Matheuristic Framework for the Chinese Postman Problem with Load-Dependent Costs*, 2026.
c: Corresponding author / PI.
https://arxiv.org/pdf/2607.22542.pdf

[**P21**] Thinh Nguyen-Cong, **Truong-Son Hy** [c], Thang N. Dinh, *Spectral Embeddings Leak Graph Topology: Theory, Benchmark, and Adaptive Reconstruction*, 2026.
c: Corresponding author / PI.
https://arxiv.org/pdf/2604.21094.pdf

[**P20**] **Truong-Son Hy**, *Q-BIOLAT: Binary Latent Protein Fitness Landscapes for QUBO-Based Optimization*, 2026.
https://arxiv.org/pdf/2603.27526.pdf

[**P19**] **Truong-Son Hy**, *Binary Latent Protein Fitness Landscapes for Quantum Annealing Optimization*, 2026.
https://arxiv.org/pdf/2603.17247.pdf

[**P18**] Brijesh FNU, Viet Thanh Duy Nguyen, Ashima Sharma, Md Harun Rashid Molla, Chengyi Xu, and **Truong-Son Hy** [c], *Multimodal Machine Learning for Soft High-k Elastomers under Data Scarcity*, 2025.
c: Corresponding author / PI.
https://arxiv.org/pdf/2601.18032.pdf

[**P17**] Kien Le Trung and **Truong-Son Hy** [c], *DiffPlace: A Conditional Diffusion Framework for Simultaneous VLSI Placement Beyond Sequential Paradigms*, 2025.
c: Corresponding author / PI.
https://arxiv.org/pdf/2510.15897.pdf

[**P16**] Phu-Vinh Nguyen, Tan-Hanh Pham, Chris Ngo, and **Truong-Son Hy** [c], *A Novel Framework for Automated Explain Vision Model Using Vision-Language Models*, 2025.
c: Corresponding author / PI.
https://www.arxiv.org/pdf/2508.20227.pdf

[**P15**] Tam Trinh, Manh Nguyen, and **Truong-Son Hy** [c], *Towards Robust Fact-Checking: A Multi-Agent System with Advanced Evidence Retrieval*, 2025.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2506.17878.pdf`

[**P14**] Nguyen Phu Vinh, Anh Chung Hoang, Chris Ngo, and **Truong-Son Hy** [c], *Repeton: Structured Bug Repair with ReAct-Guided Patch-and-Test Cycles*, 2025.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2506.08173.pdf`

[**P13**] Tan-Hanh Pham, Phu-Vinh Nguyen, Dang The Hung, Bui Trong Duong, Vu Nguyen Thanh, Chris Ngo, Tri Quang Truong, and **Truong-Son Hy** [c], *IQBench: How "Smart" Are Vision-Language Models? A Study with Human IQ Tests*, 2025.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2505.12000.pdf`

[**P12**] Tung D. Vu, Chung Hoang, and **Truong-Son Hy** [c], *Multimodal graph representation learning for website generation based on visual sketch*, 2025.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2504.18729.pdf`

[**P11**] Viet Thanh Duy Nguyen, and **Truong-Son Hy** [c], *Advances in Protein Representation Learning: Methods, Applications, and Future Directions*, 2025.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2503.16659.pdf`

[**P10**] Tuan-Anh Yang, **Truong-Son Hy** [c], and Phuong D. Dao, *MOB-GCN: A Novel Multiscale Object-Based Graph Neural Network for Hyperspectral Image Classification*, 2025.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2502.16289.pdf`

[**P9**] Cong Nga Ha, Phuc Pham, and **Truong-Son Hy** [c], *LANTERN: Leveraging Large Language Models and Transformers for Enhanced Molecular Interactions*, DOI 10.1101/2025.02.10.637522.
c: Corresponding author / PI.
`https://www.biorxiv.org/content/10.1101/2025.02.10.637522v1.full.pdf`

[**P8**] Van Quang Nguyen, Quoc Chuong Nguyen, Thu Huong Dang, and **Truong-Son Hy** [c], *Hybridising Reinforcement Learning and Heuristics for Hierarchical Directed Arc Routing Problems*, 2024.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2501.00852.pdf`

[**P7**] Khai Le-Duc, Quy-Anh Dang, Tan-Hanh Pham, and **Truong-Son Hy** [c], *wav2graph: A Framework for Supervised Learning Knowledge Graph from Speech*, 2024.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2408.04174.pdf`

[**P6**] Khai Le-Duc, Ryan Zhang, Ngoc Son Nguyen, Tan-Hanh Pham, Anh Dao, Ba Hung Ngo, Anh Totti Nguyen, and **Truong-Son Hy** [c], *LiteGPT: Large Vision-Language Model for Joint Chest X-ray Localization and Classification Task*, 2024.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2407.12064.pdf`

[**P5**] Cong Dao Tran*, Thong Bach*, and **Truong-Son Hy** *[c], *Symmetry-preserving graph attention network to solve routing problems at multiple resolutions*, 2023.
∗: Co-first authors.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2310.15543.pdf`

[**P4**] **Truong-Son Hy** and Cong Dao Tran, *Graph Attention-based Deep Reinforcement Learning for solving the Chinese Postman Problem with Load-dependent costs*, 2023.
`https://arxiv.org/pdf/2310.15516.pdf`

[**P3**] Ngoc-Dung Do*, **Truong-Son Hy***[c] and Duy Khuong Nguyen, *Sparsity exploitation via discovering graphical models in multi-variate time-series forecasting*, 2023.
∗: Co-first authors.
c: Corresponding author / PI.
`https://arxiv.org/pdf/2306.17090.pdf`

[**P2**] Erik Henning Thiede, **Truong-Son Hy** and Risi Kondor, *The general theory of permutation equivarant neural networks and higher order graph variational encoders*, 2020.
`https://arxiv.org/pdf/2004.03990.pdf`

[**P1**] **Truong-Son Hy** and Chris Jones, *Graph neural networks with efficient tensor operations in CUDA/GPU and GraphFlow deep learning framework in C++ for quantum chemistry*, 2019.
`https://hytruongson.github.io/HySonLab/CCN-GraphFlow.pdf`

PhD Students

At University of Alabama at Birmingham, I am grateful to advise talented and hard-working PhD students:

- Viet Thanh Duy Nguyen (from Spring 2025)
- Phuc Pham (from Spring 2026)

Mentoring Postdoctoral Scholars

At University of Alabama at Birmingham, it is my privilege to mentor postdoctoral scholars:

- Dr. Huu Phong Nguyen [website] (from August 1, 2026)

PhD Committee

It is my honor to serve in the PhD committee of:

- Tien Kha Pham, Deakin University, Australia in 2024. His PhD thesis' title is "Memory for Fast Adaptation in Neural Networks".
- Shishir Sarker, Department of Biology, Indiana State University, United States from 2024. His PhD thesis' theme is "Study of regulating the virulence effect by inhibiting SIP binding to RopB".

At University of Alabama at Birmingham, I am serving in the PhD committee of senior PhD students:

- Ihsan E. Buker (from 2026)
- Sandeep Chowdary Vejandla (from 2026)

I am also serving in the PhD committee (Level 1) for junior PhD students: Sharif Noor Zisad (2024), Delower Hossain (2024), Mengchen Fan (2024), Sheikh Abujar (2024), Xi Xiao (2025), Boniface Mulenga Sindala (2025).

Area Chair & Organizer

I serve as an **Area Chair (AC)** for the following top-tier conferences:

- ACL (Association for Computational Linguistics) 2025

I co-organize and serve as a senior meta-reviewer, i.e. area chair, for the following conferences:

- BIOKDD 2025 – 24th International Workshop on Data Mining in Bioinformatics, collocated with ACM SIGKDD 2025 in Toronto, Canada [website] [proceedings] [news]
- BIOKDD 2026 – 25th International Workshop on Data Mining in Bioinformatics, collocated with ACM SIGKDD 2026 in Jeju, Korea [website] [proceedings]

Editor

It is my great honor to be:

- **Reviewing Editor** for Springer Nature.
- **Editorial Board Member** for Discover Artificial Intelligence of Springer Nature.
- **Editorial Board Member** for Scientific Reports of Springer Nature.
- **Guest Editor** for IEEE/ACM Transactions on Computational Biology and Bioinformatics, Special Issue on BIOKDD 2025.
- **Handling Editor** for Frontiers in Medicine.

Reviewer

Journals (ORCID: https://orcid.org/0000-0002-5092-3757)

1. **Nature - Scientific Reports** [Certificate - 01/26/25] [Certificate - 02/24/25]
2. **Computational and Structural Biotechnology Journal** (Elsevier) [Certificate]
3. **Neural Networks** (Elsevier) [Certificate]
4. **International Journal of Biological Macromolecules** (Elsevier) [Certificate]
5. **Information Fusion** (Elsevier) [Certificate]
6. **Computer Vision and Image Understanding** (Elsevier) [Certificate]
7. **Information Processing and Management** (Elsevier) [Certificate]
8. **Knowledge-Based Systems** (Elsevier) [Certificate]
9. **Expert Systems With Applications** (Elsevier) [Certificate]
10. **Engineering Applications of Artificial Intelligence** (Elsevier) [Certificate]
11. **Applied Soft Computing** (Elsevier) [Certificate]
12. **Chemical Engineering Journal** (Elsevier) [Certificate]
13. **BMC Biology** (Springer Nature) [Certificate]
14. **BMC Bioinformatics** (Springer Nature) [Certificate]
15. **Molecular Diversity** (Springer Nature) [Certificate]
16. **Discover Artificial Intelligence** (Springer Nature) [Certificate]
17. **npj Drug Discovery** (Springer Nature) [Certificate]
18. **Quantum Machine Intelligence** (Springer Nature) [Certificate]
19. **Data Mining and Knowledge Discovery** (Springer Nature) [Certificate]
20. **Artificial Intelligence In Medicine** (Springer Nature) [Certificate]
21. **Journal of Cheminformatics** (Springer Nature) [Certificate]
22. **Nature - Communications Biology** (Springer Nature)
23. **Nature - Communications** (Springer Nature)
24. **Machine Learning** (Springer Nature)
25. **PLOS Computational Biology**
26. **SIAM Journal on Matrix Analysis and Applications**
27. **IEEE Journal of Biomedical and Health Informatics**
28. **IEEE/ACM Transactions on Computational Biology and Bioinformatics** [Certificate 2024] [Certificate 2025]
29. **IEEE Transactions on Multimedia**
30. **IEEE Transactions on Mobile Computing** [Certificate 2025]
31. **IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems**
32. **ACM Transactions on Knowledge Discovery from Data**
33. **ACM Transactions on Intelligent Systems and Technology**
34. **National Science Review** (Oxford University Press)
35. **Biology Methods and Protocols** (Oxford University Press)
36. **Advanced Science** (Wiley)
37. **ChemMedChem** (Wiley) [Certificate]
38. **Angewandte Chemie International Edition** (Wiley)
39. **Journal of Chemical Information and Modeling** (American Chemical Society) [Certificate 2025]
40. **ACS Omega** (American Chemical Society)
41. **Frontiers in Artificial Intelligence** (Frontiers) [Certificate]
42. **Frontiers in Public Health** (Frontiers) [Certificate]
43. **Machine Learning: Science and Technology** (IOP Publishing)
44. **Machine Learning: Health** (IOP Publishing)

45. **Physica Scripta** (IOP Publishing)
46. **Applied Artificial Intelligence** (Taylor & Francis) [Certificate]
47. **Journal of Machine Learning Research (JMLR)**

Conferences

1. **NeurIPS:** 2026, 2025, 2024, 2023, 2022, 2021
2. **CVPR:** 2026, 2025
3. **ICCV:** 2025
4. **WACV:** 2026
5. **ICML:** 2026, 2022
6. **ICLR:** 2026, 2025, 2024
7. **AISTATS:** 2026, 2025, 2024
8. **AAAI:** 2026, 2025, 2024, 2023
9. **KDD:** 2025, 2024
10. **BIO KDD:** 2025
11. **Learning on Graphs (LoG):** 2023, 2022
12. **EMNLP:** 2025, 2023
13. **ACL:** 2025, 2024
14. **NAACL:** 2025
15. **IEEE ROMAN:** 2024
16. **ICASSP:** 2025 [Certificate]

Service

- **Judge**, ICPC North America South Division Regional Contest, 2026.
- **Coach**, University of Alabama at Birmingham (UAB) ICPC Team, 2025 [Certificate].
- **Co-organizer and Problem Setter**, Alabama High School Programming Contest, 2025 [Certificate], 2026 [Certificate].

Software [HySonLab] [HyTruongSon]

50. MutaStruct: Revisiting Benchmarks for Single-Site Mutation-Induced Structural Change Prediction.
`https://github.com/HySonLab/MutaStruct`

49. AMP-atlas: A Browser-Only Workbench for Antimicrobial Peptide Exploration and Interactive Helical-Wheel Design.
`https://hysonlab.github.io/amp-atlas/` (website)
`https://github.com/HySonLab/amp-atlas` (code)
`https://arxiv.org/pdf/2610.29096.pdf` (paper)

48. PRIME-ISP: Towards a Unified Framework for Protein Interaction Site Prediction Using Hierarchical Protein Representations.
`https://github.com/HySonLab/PRIME-ISP`

47. PRIME: Protein Representation via Physics-Informed Multiscale Equivariant Hierarchies.
`https://github.com/HySonLab/PRIME`

46. Q-BIOLAT: Binary Latent Protein Fitness Landscapes for Quantum Annealing Optimization.
`https://github.com/HySonLab/Q-BIOLAT`
`https://github.com/HySonLab/Q-BIOLAT-Extended` (Extended Repository)

45. AgentIoT: Self-Evolving Multi-Agent Network for Industrial IoT Predictive Maintenance.
`https://github.com/HySonLab/AgentIoT`

44. Polymers: Multimodal Machine Learning for Soft High-k, Low-Modulus Polymers under Data Scarcity.
`https://github.com/HySonLab/Polymers`

43. TeBaAb: Text-Based Antigen-Conditioned Antibody Redesign via Directed Evolution.
https://github.com/HySonLab/TeBaAb

42. LINKER: Learning Interactions Between Functional Groups and Residues With Chemical Knowledge-Enhanced Reasoning and Explainability.
https://github.com/HySonLab/LINKER

41. EvoSynth: Multi-Target Drug Discovery through Latent Evolutionary Optimization and Synthesis-Aware Prioritization.
https://github.com/HySonLab/EvoSynth

40. AdaptFNO: Adaptive Fourier Neural Operator with Dynamic Spectral Modes and Multi-scale Learning.
https://github.com/HySonLab/AdaptFNO

39. DiffPlace: A Conditional Diffusion Framework for Simultaneous VLSI Placement Beyond Sequential Paradigms.
https://github.com/HySonLab/DiffPlace/

38. FactAgent: A sophisticated multi-agent fact-checking system that combines advanced evidence retrieval techniques to verify factual claims across diverse domains.
https://github.com/HySonLab/FactAgent

37. MatCPP: Efficient Matheuristics for Solving the Chinese Postman Problem with Load Constraints.
https://github.com/HySonLab/MatCPP

36. PointDistill: Topology-Guided Knowledge Distillation for Efficient Point Cloud Processing.
https://github.com/HySonLab/PointDistill

35. EquiHGNN: Rotationally Equivariant HyperGraph Neural Networks.
https://github.com/HySonLab/EquiHGNN/

34. Design2Code: Multimodal graph representation learning for website source code generation given visual sketch.
https://github.com/HySonLab/Design2Code **33. WaveletPE:** Range-aware Graph Positional Encoding via High-order Pretraining.
https://github.com/HySonLab/WaveletPE

32. LANTERN: Leveraging Large Language Models and Transformers for Enhanced Molecular Interactions.
https://github.com/HySonLab/LANTERN

31. MOB-GCN: Multiscale Object-Based Graph Neural Network for Hyperspectral Image Segmentation and Classification.
https://github.com/HySonLab/MultiscaleHSI

30. BioMedKG: Multimodal Contrastive Representation Learning in Augmented Biomedical Knowledge Graphs.
https://github.com/HySonLab/BioMedKG

29. ArcRoute: Hybrid algorithm combining Reinforcement Learning (RL) and heuristics to solve Hierarchical Directed Capacitated Arc Routing Problem (HDCARP).
https://github.com/HySonLab/ArcRoute

28. DrugPipe: Generative AI-assisted Drug Repurposing Pipeline.
https://github.com/HySonLab/DrugPipe

27. TESGNN: 3D Temporal Equivariant Scene Graph Neural Networks.
https://github.com/HySonLab/TESGraph

26. LightMed: A Light-weight and Robust FFT-Based Model for Adversarially Resilient Medical Image Segmentation.
https://github.com/HySonLab/LightMed

25. EntityKG: A Framework for Supervised Learning Knowledge Graph from Speech.
https://github.com/HySonLab/EntityKG

24. VietMedKG: Knowledge Graph and Benchmark for Traditional Vietnamese Medicine.
https://github.com/HySonLab/VietMedKG

23. ML4Insects: A library for EPG signal analysis of pierce-sucking insects.
https://github.com/HySonLab/ML4Insects/

22. Hierarchical Attention: Scalable Hierarchical Self-Attention with Learnable Hierarchy for Long-Range Interactions (TMLR).
https://github.com/HySonLab/HierAttention

21. EquiMesh: E(3)-Equivariant Mesh Neural Networks (AISTATS 2024).
https://github.com/HySonLab/EquiMesh

20. Protein-Redesign: Complex-based Ligand-Binding Proteins Redesign by Equivariant Diffusion-based Generative Models.
https://github.com/HySonLab/Protein_Redesign

19. LatentDE: Latent-based Directed Evolution for Protein Design.
https://github.com/HySonLab/LatentDE

18. Protein-Pretrain: Multimodal Pretraining for Unsupervised Protein Representation Learning.
https://github.com/HySonLab/Protein_Pretrain

17. Protein-Design: Protein Design by Machine Learning guided Directed Evolution.
https://github.com/HySonLab/Directed_Evolution

16. Multires-NP-hard: Symmetry-preserving and multiresolution Reinforcement Learning to solve NP-hard problems in Operations Research including TSP and VRP.
https://github.com/HySonLab/Multires-NP-hard

15. CPP-LC: Implementation in C++ for the Chinese postman problem with load constraints (CPP-LC) with Evolutionary Algorithm, Ant Colony Optimization and many other meta-heuristics.
https://github.com/HySonLab/Chinese_Postman_Problem

14. Ligand generation: Target-aware Variational Auto-encoders for Ligand Generation with Multimodal Protein Representation Learning.
https://github.com/HySonLab/Ligand_Generation

13. GraphLASSO: Sparsity exploitation via discovering graphical models in multi-variate time-series forecasting.
https://github.com/HySonLab/GraphLASSO

12. Machine Learning for Epidemiology: Predicting COVID-19 pandemic by spatio-temporal graph neural networks.
https://github.com/HySonLab/pandemic_tgnn

11. Multiresolution Graph Transformers: Learning hierarchical structures including proteins and polymers by multiresolution graph transformers and wavelet positional encoding.
https://github.com/HySonLab/Multires-Graph-Transformer

10. TWGNN: Fast Temporal Wavelet Graph Neural Networks for learning timeseries with underlying graph structure with applications in traffic prediction and brain networks.
https://github.com/HySonLab/TWGNN

9. ViDeBERTa: A powerful pre-trained language model for Vietnamese (EACL 2023).
`https://github.com/HySonLab/ViDeBERTa`

8. Drugs-Proteins knowledge graph: Large-scale deep generative models on multi-modal knowledge graph for predicting drug interactions.
`https://github.com/HySonLab/drug-interactions`

7. Spherical CNNs: PyTorch implementation of Spherical Convolutional Neural Networks with Clebsch–Gordan transform for nonlinearity in the Fourier space.
`https://github.com/risilab/SphericalNet`

6. Learnable MMF: Learning Multiresolution Matrix Factorization and its Wavelet Networks on Graphs.
`https://github.com/risilab/Learnable_MMF`

5. MGVAE: Multiresolution Equivariant Graph Variational Autoencoder (MGVAE) and Multiresolution Graph Networks (MGN) for supervised molecular properties prediction, unsupervised molecular representation learning, graph generation, citation link prediction and graph-based image generation.
`https://github.com/HyTruongSon/MGVAE`

4. LibCCNs: Covariant Compositional Networks Library is an easy-to-use and efficient implementation of Covariant Compositional Networks (CCNs) with TensorFlow and PyTorch's APIs based on a shared common C++ core.
`https://github.com/HyTruongSon/LibCCNs`

3. Invariant Graph Networks: A PyTorch implementation of the Invariant and Equivariant Graph Networks.
`https://github.com/HyTruongSon/InvariantGraphNetworks-PyTorch`

2. GraphFlow: Deep Learning framework built from scratch in C++/CUDA that supports symbolic/automatic differentiation, dynamic computation graphs, tensor/matrix operations accelerated by GPU and implementations of various state-of-the-art graph neural networks and other Machine Learning models.
`https://github.com/HyTruongSon/GraphFlow`

1. Fourier Transform Library: 1D/2D FFT, DFT and DCT; JPEG image compression.
`https://github.com/HyTruongSon/Fourier-Transform-Library`

Media

- UAB wrote an article about HySonLab's EVOSYNTH framework published at Nature's Communications Chemistry: [link]
- Interview with Voice of Vietnam (VOV): [link]
- The newspaper of Vietnam Ministry of Science & Technology wrote an article about my lab's research in AI for Protein Design: [link]
- UAB wrote an article about my lab's research in AI for Drug Discovery: [link]
- Article about myself at Thanh Nien Newspaper, one of the major newspapers in Vietnam: [Vietnamese] [English]
- Article about my advices for younger generation at Thanh Nien Newspaper: [Vietnamese]
- At my middle school in 2024: [link]
- At UChicago in 2020: [link]
- At ELTE in 2016: [link]

Invited Talks

20. Viettel High Tech June 30, 2026
Talk: *Generative AI and Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Dr. Nam Nguyen.

19. VNU University of Engineering and Technology June 18, 2026
Talk: *Generative AI and Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Dr. Le Duc Trong.

18. Military Technical Academy, Vietnam [news] [flyer] June 11, 2026
Talk: *Generative AI and Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Dr. Hy Duc Manh.

17. DBIDS PowerTalk Seminar Series [flyer] September 26, 2025
Talk: *Generative AI and Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Department of Biomedical Informatics and Data Science, UAB School of Medicine.

16. Adobe Research August 28, 2025
Talk: *Geometric Deep Learning: From 3D Point Clouds and Meshes to Scene Understanding and Distillation.*
Host: Dr. Viet Lai.

15. Universitas Gadjah Mada, Indonesia [certificate] July 29, 2025
Talk: *Generative AI & Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Prof. Muhammad Husni Santriaji.

14. Robotics & AI Expert Talk [certificate] July 18, 2025
Talk: *Generative AI & Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Dr. Tan-Hanh Pham, Harvard Medical School.

13. UAB Department of Mechanical and Materials Engineering January 24, 2025
Talk: *Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Prof. Mark Banaszak Holl.

12. University of Alabama at Birmingham May 15, 2024
Talk: *Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Prof. Ragib Hasan.

11. University of Massachusetts Boston April 11, 2024
Talk: *Geometric Deep Learning for Protein Science and Drug Discovery.*
Host: Prof. Albert Kao.

10. VinUniversity December 28, 2023
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: Prof. Le Duy Dung (Andrew).

9. Indiana State University February 1, 2023
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: Prof. Arash Rafiey.

8. Virginia Commonwealth University September 30, 2022
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: Prof. Thang Dinh.

7. Le Quy Don Technical University (Military Technical Academy) September 8, 2022
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: Faculty of Information Technology.

6. FPT Software June 2, 2022
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: FPT AI Center.

5. Vector Institute for Artificial Intelligence April 6, 2022
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: Prof. Pascal Poupart.

4. University of California San Diego February 24, 2022
Talk: *Graph representation learning and Multiresolution machine learning.*
Host: Prof. Rose Yu, Prof. Yusu Wang.

3. California Institute of Technology February 3, 2022
Talk: *Graph representation learning, deep generative models on graphs and multiresolution ma-*

chine learning.
Host: Prof. Anima Anandkumar.

2. University of Illinois Chicago January 26, 2022
Talk: *Graph representation learning and Deep generative models on graphs.*
Host: Department of Mathematics, Statistics, and Computer Science.

1. Argonne National Laboratory January 5, 2022
Talk: *Graph representation learning, deep generative models on graphs and multiresolution machine learning.*
Host: Dr. Stefan Wild, Dr. Emil Constantinescu.

Senior Collaborators

- Prof. Risi Kondor, University of Chicago [website]
- Prof. Yusu Wang, University of California San Diego [website]
- Prof. Andrey Rzhetsky, University of Chicago [website]
- Prof. Siamak Ravanbakhsh, McGill University / MILA (Canada) [website]
- Prof. Long Tran-Thanh, University of Warwick (United Kingdom) [website]
- Prof. Huong Dang, Lancaster University (United Kingdom) [website]
- Prof. Erik Thiede, Cornell University [website]
- Prof. Arash Rafiey, Indiana State University [website]
- Prof. Takuya Akiyama, Indiana State University [website]
- Prof. Kyu Hong Cho, Indiana State University [website]
- Prof. Thang N. Dinh, Virginia Commonwealth University [website]
- Prof. Phuong Dao, Colorado State University [website]
- Prof. Vamsi Nalam, Colorado State University [website]
- Prof. Rhett Davis, North Carolina State University [website]
- Prof. Nhung Nghiem, Australian National University (Australia) [website]
- Dr. Thieu N. Vo, Ton Duc Thang University (Vietnam) [website]
- Dr. Nghi D. Q. Bui, FPT Software AI Center (Vietnam) [website]
- Dr. Dao Huu Hung, FPT Software AI Center (Vietnam) [website]
- Dr. Van Nguyen, FPT Software AI Center (Vietnam) [website]
- Dr. Rajeev Jain, Qualcomm [website]
- Dr. Michael Defferrard, Qualcomm [website]
- Prof. Matthew Might, University of Alabama at Birmingham [website]
- Prof. Jake Chen, University of Alabama at Birmingham [website]

Teaching Assistant (UChicago)

9. CAPP 30271 - Mathematics for Computer Science and Data Analysis Winter 2022
Instructor: Prof. Amitabh Chaudhary

8. MPCS 53112 - Advanced Data Analytics Autumn 2021
Instructor: Prof. Amitabh Chaudhary

7. CMSC 25025 - Machine Learning & Large-Scale Data Analysis Spring 2021
Instructor: Prof. Yali Amit

6. CMSC 35400 - Machine Learning Winter 2021
Instructor: Prof. Yuxin Chen
Website: `https://sites.google.com/uchicago.edu/stat-37710-cmsc-35400-w21/`

5. CMSC 25025 - Machine Learning & Large-Scale Data Analysis Spring 2019
Instructor: Prof. Yali Amit

4. CMSC 15100 - Introduction to Computer Science I Autumn 2018
Instructor: Adam Shaw & Matthew Wachs
Website: `http://people.cs.uchicago.edu/~adamshaw/cmsc15100-2018/index.html`

3. MPCS 53111 - Machine Learning Spring 2017
Instructor: Prof. Amitabh Chaudhary
Website: `https://mpcs-courses.cs.uchicago.edu/2015-16/spring/courses/53111`

2. CMSC 25400 - Machine Learning Winter 2017
Instructor: Prof. Risi Kondor
Website: `http://people.cs.uchicago.edu/~risi/cmsc25400.html`

1. CMSC 22600 - Compilers for Computer Languages Autumn 2016
Instructor: Prof. John Reppy
Website: `https://www.classes.cs.uchicago.edu/archive/2016/fall/22600-1/`

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE): Since 2024 [IEEE Author Profile]
- Association for Computing Machinery (ACM): Since 2024
  [ACM Author Profile 1] [ACM Author Profile 2] [ACM Author Profile 3]

Qualifications

Programming Languages: C/C++, Java, Python, Matlab, Haskell, Ada, Pascal, SQL, HTML/CSS.
Libraries: TensorFlow, PyTorch, STL, OpenGL, LaTeX.
Tools: Vim, Netbeans, Eclipse, Codeblocks, Dev-C++, Microsoft Visual Studio & Office.